

BUA CHUM, Thailand

WMO: 484180

Lat: 15.267N

Lon: 101.187E

Elev: 169

StdP: 14.61

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	59.8	62.6	49.4	52.5	68.4	52.6	59.2	69.7	18.7	77.9	17.8	78.6	0.9	50	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.3	101.3	79.5	99.4	79.2	97.6	79.0	82.8	93.4	82.1	92.4	81.6	91.7	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	158.4	87.4	79.8	155.4	86.9	79.2	152.3	86.3	46.9	93.9	46.2	92.6	45.6	92.0	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.7	10.8	DB	55.5	103.9	4.0	2.0	52.6	105.4	50.3	106.6	48.1	107.7	45.2	109.2	(4)
(5)			WB	52.1	84.5	3.4	1.1	49.6	85.3	47.6	86.0	45.6	86.6	43.1	87.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	77.9	82.0	86.0	87.7	86.6	85.8	84.5	83.9	83.1	82.6	80.2	77.4
	DBStd	4.49	4.15	4.15	3.78	3.33	2.96	2.35	2.14	1.88	1.97	2.01	3.45	4.38
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD50	12100	866	896	1115	1130	1135	1074	1070	1052	994	1012	907	849
	CDD65	6627	402	476	650	680	670	624	605	587	544	547	457	385
	CDH74	75098	3794	5348	8338	8873	8265	7699	7013	6487	5713	5678	4407	3483
CDH80	35377	1665	2758	4633	4987	4186	3745	3115	2694	2164	2203	1816	1411	
Wind	WSAvg	2.5	3.3	3.2	3.1	2.5	2.1	2.2	2.2	1.8	1.2	1.8	2.7	3.6
Precipitation	PrecAvg	45.0	0.4	0.3	1.9	3.2	5.6	5.2	5.0	7.2	10.2	5.0	0.8	0.2
	PrecMax	58.1	1.9	2.4	5.6	7.7	9.1	9.3	10.3	11.5	15.9	14.3	6.3	2.6
	PrecMin	32.8	0.0	0.0	0.0	0.1	1.5	1.3	2.3	4.2	6.0	0.0	0.0	0.0
	PrecStd	6.3	0.6	0.5	1.5	1.7	1.8	2.1	1.9	1.9	2.7	3.0	1.3	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.5	100.1	103.1	104.2	102.5	99.9	97.5	95.4	94.2	94.3	95.6	94.7
		MCWB	75.2	77.3	77.8	80.1	81.8	80.8	80.9	80.9	81.8	80.5	78.4	76.8
	2%	DB	93.4	97.6	100.5	101.8	99.2	97.0	94.9	93.5	92.4	92.3	93.0	92.3
		MCWB	74.9	76.5	78.0	79.8	80.9	80.4	80.3	80.6	81.2	79.7	77.6	75.6
	5%	DB	91.4	95.6	98.6	99.4	96.7	94.8	93.0	91.9	90.9	90.9	91.0	90.0
		MCWB	74.4	76.0	78.0	79.7	80.6	80.5	80.0	80.4	80.6	79.0	76.9	74.5
	10%	DB	88.8	93.2	96.5	97.2	94.2	93.0	91.3	90.4	89.3	89.3	88.9	87.6
		MCWB	73.1	75.6	77.8	79.5	80.5	80.1	79.7	80.1	80.2	78.6	75.8	73.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.1	80.0	81.5	83.5	83.6	82.8	82.7	82.8	83.2	82.6	81.6	79.7
		MCDB	90.3	94.5	96.4	98.0	96.0	93.4	92.2	91.2	91.4	90.7	91.2	90.0
	2%	WB	77.6	78.9	80.4	82.0	82.6	81.9	81.7	81.9	82.1	81.5	80.2	77.9
		MCDB	88.6	93.0	94.7	95.9	94.4	92.3	91.0	90.3	89.8	89.4	89.0	88.0
	5%	WB	76.3	78.0	79.6	81.2	81.8	81.4	80.9	81.2	81.3	80.7	79.1	76.4
		MCDB	86.7	91.3	93.5	94.7	92.9	91.5	90.1	89.5	88.8	88.4	87.0	86.3
	10%	WB	75.0	77.0	78.9	80.5	81.0	80.8	80.3	80.6	80.7	80.0	77.8	75.0
		MCDB	84.8	89.3	92.4	93.5	91.4	90.6	89.1	88.7	87.9	86.8	85.0	84.7
Mean Daily Temperature Range	5% DB	MDBR	22.1	22.6	20.9	19.3	16.3	15.3	14.4	14.0	13.0	13.9	17.3	20.3
		MCDBR	25.1	25.5	23.7	21.9	19.4	17.6	16.6	16.1	15.2	16.1	20.0	22.6
		MCWBR	9.1	8.1	6.6	6.2	5.6	5.0	5.1	5.5	5.7	5.5	6.7	8.0
	5% WB	MCDBR	21.0	22.7	21.3	20.0	17.4	16.0	15.2	15.1	14.2	14.7	16.8	20.7
MCWBR		7.9	7.3	6.4	6.3	6.0	5.5	5.6	5.9	5.9	5.7	6.3	8.1	
Clear-Sky Solar Irradiance	taub	0.505	0.602	0.657	0.602	0.507	0.458	0.447	0.452	0.490	0.541	0.485	0.478	
	taud	2.024	1.809	1.718	1.883	2.190	2.334	2.363	2.333	2.210	2.028	2.167	2.124	
	Ebn at Noon	249	231	222	235	255	265	268	270	260	243	254	253	
	Edh at Noon	53	69	77	65	47	41	39	41	46	54	46	47	
All-Sky Solar Radiation	RadAvg	1540	1668	1728	1862	1776	1679	1588	1544	1494	1532	1552	1517	
	RadStd	102	67	97	78	102	92	85	93	81	95	75	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	+0.47	+0.50	N/A	N/A	+180	+176

Nomenclature: See separate page